

Algebra 1
Unit 2
Lesson 6 Homework

Name Answer Key
Date _____
Period _____

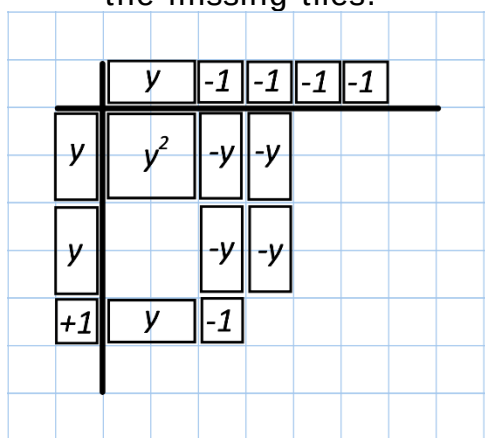
1. Model the following expression with algebra tiles.

$$5x - 4$$

x	x	x	x	x	-1	-1	-1	-1
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Use the expression $(2y + 1)(y - 4)$ for questions 2 – 4.

2. Complete the model to find the product of $(2y + 1)(y - 4)$ by drawing and labeling the missing tiles.



	y	-1	-1	-1	-1
y	y^2	$-y$	$-y$	$-y$	$-y$
y	y^2	$-y$	$-y$	$-y$	$-y$
+1	y	-1	-1	-1	-1

3. Determine how many of each tile are represented in the product and write the numbers in the table.

y^2 tiles	$+y$ tiles	$-y$ tiles	$+1$ tiles	-1 tiles
2	1	8	0	4

4. Complete the statement.

The product of $(2y + 1)(y - 4)$ is $2y^2 - 7y - 4$

For questions 5-9, use algebra tiles to find each product.

5. $(4 - 3x)(x - 1) = -3x^2 + 7x - 4$

	-x	-x	-x	1	1	1	1
x	$-x^2$	$-x^2$	$-x^2$	x	x	x	x
-1	x	x	x	-1	-1	-1	-1

6. $(2x - 1)^2 = 4x^2 - 4x + 1$

	x	x	-1
x	x^2	x^2	$-x$
x	x^2	x^2	$-x$
-1	$-x$	$-x$	1

7. $(x - 3)(x + 3) = x^2 - 9$

	x	-1	-1	-1
x	x^2	$-x$	$-x$	$-x$
+1	x	-1	-1	-1
+1	x	-1	-1	-1
+1	x	-1	-1	-1

8. $(x - 5)(x + 1) = x^2 - 4x - 5$

	x	-1	-1	-1	-1	-1
x	x^2	$-x$	$-x$	$-x$	$-x$	$-x$
1	x	-1	-1	-1	-1	-1

9. $(1 - x)(2x + 2) = -2x^2 + 2$

	-x	+1
x	$-x^2$	x
x	$-x^2$	x
+1	$-x$	+1
+1	$-x$	+1

10. One of your classmates was absent when you learned how to multiply polynomials using algebra tiles. Write them a note explaining the process.

Multiplying polynomials using algebra tiles is just like finding the area of a rectangle. Line-up the factors to represent length and width. Fill in the area with blocks that are the same length and width. The blocks in the area space represent the product of the polynomials.